

Task 1 - Data Upload & Quality Assessment

Part A - Upload + Quality Assessment Prompt

I have a weekly sales CSV with order, date, region, product, category, quantity, price, revenue, sales rep and status. Act as a data analyst. First describe the dataset structure, including row count, column count and what each column means. Then identify all data-quality problems before analysis. Check missing values, inconsistent formats/casing, non-numeric values and unusual outliers. Do not change the data yet. Return a short structure summary followed by a quality-issues list.

Rows and columns

20 data rows and 10 columns.

All data quality issues found

Inconsistent date format in order 1003; mixed casing in rep_name; lowercase 'pending' in order 1010; missing qty and unit_price in order 1004; missing qty in order 1020; revenue is N/A in order 1010; qty 99 in order 1019 is a strong outlier. The pending rows also contain zero/missing sales values that should be reviewed before analysis.

Issues AI missed

None of the deliberately planted issues were missed in this review.

Did AI describe the data correctly?

Yes. It represents weekly sales/order records with products, regions, quantities, prices, revenue, sales representatives and order status.

Developer validation script request prompt

Write a Python pandas validation script for sales.csv. Load the CSV, print missing-value counts by column, detect inconsistent casing in rep_name and status, identify rows where revenue is non-numeric, flag qty values above 50, and print a clear validation report. Do not silently fix the data.

Script output summary

The script loads the CSV, checks nulls, compares text values with normalized Title Case, converts revenue with errors='coerce' to find invalid values, flags qty > 50, and prints the affected rows and counts.